

PART 2 – INTERVIEW & CAREER INTELLIGENCE ENGINE

7. Interview Simulation Engine

7.1 Role Selection

Student selects:

- Target job role
- Skill stack

System validates resume-role alignment.

7.2 Role-Based Question Generator

Mapping Logic:

Role × Skill × Category × Difficulty

Categories:

- Technical
- Behavioral
- System Design
- Project Deep Dive

Difficulty Levels:

- Beginner
- Intermediate
- Advanced

7.3 Interview Session Lifecycle

1. Role selected
2. Questions generated
3. Student answers submitted
4. Evaluation triggered
5. Report generated
6. Session saved

7.4 Answer Evaluation Engine

Each answer scored on:

- Keyword Relevance – 30%
- Technical Depth – 30%
- Logical Structure – 20%
- Terminology Usage – 10%
- Completeness – 10%

Outputs:

- Score (0–100)
- Missing concepts
- Actionable feedback

Evaluation latency target: < 3 seconds

8. Unified Career Intelligence & Optimization Engine

This module integrates interview performance, communication evaluation, readiness scoring, and adaptive feedback.

It is the decision-making core of the platform.

8.1 Interview Readiness Score (IRS)

Formula:

IRS =

$(\text{Resume Strength} \times 0.20) +$
 $(\text{Technical Performance} \times 0.40) +$
 $(\text{Behavioral Performance} \times 0.20) +$
 $(\text{Role Skill Match} \times 0.20)$

Classification:

85–100 → Highly Ready

70–84 → Moderately Ready

50–69 → Developing

Below 50 → Needs Significant Improvement

8.2 Communication Clarity Index (CCI)

Evaluates:

- Grammar accuracy
- Logical sequencing
- Concept articulation
- Redundancy detection
- STAR method compliance

Score Range:

0–100

Classification:

80+ → Excellent

60–79 → Good

40–59 → Fair

Below 40 → Needs Improvement

8.3 Career Readiness Score (CRS)

CRS Formula:

CRS =

$(\text{Learning Mastery} \times 0.30) +$
 $(\text{Interview Readiness} \times 0.40) +$
 $(\text{Consistency Score} \times 0.10) +$
 $(\text{Role Alignment} \times 0.20)$

CRS represents overall placement preparedness.

8.4 Intelligent Feedback & Adaptive Optimization Loop

After interview session:

System performs cross-analysis between:

- Weak interview responses
- Topic mastery scores
- Resume skill gaps

If weakness detected:

- Flag related topics as High Priority
- Inject into study plan
- Increase revision frequency

- Recommend targeted mock interviews

Closed Loop:

Learn → Test → Evaluate → Adjust → Improve

8.5 Growth Tracking & Trend Analysis

For every interview attempt, system stores:

- Interview Readiness Score
- Technical Score
- Behavioral Score
- Communication Clarity Score
- Career Readiness Score
- Timestamp

Dashboard Visualizations:

- Readiness trend over time
- Skill gap evolution
- Improvement percentage
- Performance stability